

SpaceX Falcon 9 Landing Prediction

Detailed repository report | IBM Applied Data Science Capstone

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Record type	Educational software and data-science capstone
Version	v1.0.0
DOI	10.5281/zenodo.22763763
Report date	15 September 2026

Executive summary

This report documents an educational end-to-end data-science repository developed for the IBM Applied Data Science Capstone within the IBM Data Science Professional Certificate on Coursera. The project collects and prepares historical SpaceX Falcon 9 launch data, explores factors associated with first-stage landing outcomes, develops interactive visualisations and compares supervised classification models. It is a portfolio and training project rather than an operational aerospace forecasting system or peer-reviewed research study.

1 Project purpose and background

Falcon 9 first-stage recovery is central to launch-vehicle reuse. The capstone frames landing outcome as a supervised classification problem and demonstrates how API data, web-scraped tables, exploratory analysis, geospatial visualisation, dashboard development and machine learning can be combined in one reproducible workflow. The goal is to predict whether the first stage lands successfully and to examine how launch site, payload mass, orbit and launch history relate to observed outcomes.

The project was completed as coursework using the IBM Skills Network project framework and public data sources. Results should therefore be interpreted as evidence of applied data-science competence, not as an independently validated operational model.

2 Methods and software

Stage	Methods and software
Data acquisition	SpaceX REST API requests and Wikipedia table extraction using Requests and BeautifulSoup.
Preparation	Pandas and NumPy for cleaning, missing-value handling, transformation and feature engineering.
Exploratory analysis	Pandas, Matplotlib, Seaborn, Plotly and SQL/SQLite queries to examine landing outcomes across launch sites, orbit types, payload ranges and time.
Geospatial analysis	Folium and OpenStreetMap to map launch sites, outcomes and geographic distances.
Interactive dashboard	Plotly Dash application with launch-site selection and payload-range filtering.
Machine learning	Scikit-learn pipelines comparing logistic regression, support vector machine, decision tree and k-nearest neighbours; GridSearchCV for hyperparameter optimisation.
Evaluation	Cross-validation, held-out test accuracy and confusion-matrix inspection.
Environment	Python 3.12, Jupyter Notebook, Visual Studio Code and pip requirements.

3 Repository structure

File or directory	Role
01_SpaceX_Data_Collection_API.ipynb	Retrieves and prepares launch data from the SpaceX API.
02_SpaceX_Web_Scraping.ipynb	Extracts supplementary launch information from Wikipedia.
03_SpaceX_Data_Wrangling.ipynb	Cleans, transforms and prepares modelling variables.
04_SpaceX_EDA_Data_Visualization.ipynb	Exploratory statistical and graphical analysis.
05_SpaceX_EDA_SQL.ipynb	SQL-based interrogation of the launch dataset.
06_SpaceX_Folium_Launch_Site_Analysis.ipynb	Interactive mapping and geographic analysis.
07_SpaceX-Machine-Learning-Prediction.ipynb	Model training, tuning, comparison and evaluation.
08_SpaceX_Dash_Application.py	Interactive Plotly Dash application.
datasets, database and HTML	Processed CSV files, SQLite database and exported interactive map.
images / requirements.txt	Preview figures and reproducible Python dependency list.

4 Main analyses and results

- The workflow assembled launch records through API collection and web scraping, then produced cleaned datasets for analysis and modelling.
- Exploratory analyses examined changes in landing success over time and relationships involving launch site, payload mass, orbit and booster characteristics.
- The Folium notebook mapped launch-site locations and supported geographic distance analysis; the Dash application enabled interactive site and payload filtering.
- Four supervised classifiers were compared after hyperparameter tuning: logistic regression, support vector machine, decision tree and k-nearest neighbours.

- The repository reports the support vector machine as the best-performing classifier on the held-out test set, with an accuracy of 83.3%. This is a project-specific estimate and should not be interpreted as external validation.

The project demonstrates integration of data acquisition, SQL, visual analytics and machine learning within a coherent portfolio workflow. The principal result is methodological: a reproducible capstone showing how multiple data-science components can be connected from raw public data to model evaluation and interactive communication.

5 Limitations

- The work is an educational capstone based on an IBM Skills Network framework, not an independently designed aerospace study or peer-reviewed publication.
- The dataset is historical and relatively small; a single held-out test accuracy can be sensitive to the chosen split and class composition.
- The reported 83.3% accuracy is not evidence of performance on future launches, changing vehicle configurations or external datasets.
- Accuracy alone may conceal class-specific errors; precision, recall, F1 score, calibration and uncertainty would strengthen evaluation.
- Associations involving site, payload or orbit may reflect correlated mission characteristics and should not be interpreted causally.
- API and Wikipedia content can change. Re-running data acquisition later may not reproduce the exact archived datasets without fixed snapshots.
- GridSearchCV on a small dataset can yield unstable selections; repeated or nested resampling would provide a stronger estimate of generalisation.
- The dashboard is a local demonstration and is not presented as a continuously maintained production service.

6 Reproducibility instructions

Clone the repository, create an isolated Python environment and install the declared dependencies.

```
git clone https://github.com/dalalalhabad/IBM-Applied-Data-Science-Capstone.git
cd IBM-Applied-Data-Science-Capstone
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
jupyter notebook
```

- Open notebooks 01 through 07 in numerical order because later stages rely on outputs produced by earlier stages.
- For an exact archived reproduction, retain the supplied CSV and SQLite files. API and web-scraping reruns may return changed records.
- Record the Python and package versions used for any new run. The README specifies Python 3.12 and the required libraries are listed in requirements.txt.
- Run the dashboard with `python 08_Spacex_Dash_Application.py` and use the local address printed by Dash.
- Compare regenerated figures and model metrics with the archived repository outputs. Document any differences due to data-source or package-version changes.

- For future model claims, freeze the dataset, preserve the train/test split or random seeds, and report additional class-specific metrics.

7 Access and persistent identifier

GitHub: github.com/dalalalhabad/IBM-Applied-Data-Science-Capstone

Zenodo DOI: [10.5281/zenodo.22763763](https://doi.org/10.5281/zenodo.22763763)

Primary data sources: SpaceX REST API and the Wikipedia list of Falcon 9 and Falcon Heavy launches, as documented in the repository.

8 Recommended citation

Alhabad, D. F. S. A. (2026). SpaceX Falcon 9 Landing Prediction (Version 1.0.0) [Computer software; IBM Applied Data Science Capstone]. Zenodo. <https://doi.org/10.5281/zenodo.22763763>

The IBM Skills Network and Coursera should be acknowledged as the educational source of the capstone framework. Data users should also cite the underlying SpaceX API and Wikipedia sources where appropriate.

9 Licence and reuse

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Prepared from the repository documentation and release record current on 15 September 2026.